

NTIRE 2026 Image Shadow Removal Challenge Factsheet

LiteRestormer: Lightweight Residual CNN for Efficient Shadow Removal

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1. Team Details

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- **Best Scoring Entry (Test):** PSNR: 19.889 dB

2. Method Details

2.1. General Method Description

The proposed method, *LiteRestormer*, is a lightweight convolutional neural network (CNN) designed to eliminate cast shadows while preserving original scene structure. The architecture prioritizes computational efficiency, achieving a runtime of approximately 0.43 seconds per image on a GPU. The pipeline consists of three primary stages:

1. **Feature Expansion:** A single 3×3 convolution expands the 3-channel RGB image into a 32-channel feature space to recognize complex textures and shadow boundaries.
2. **Residual Processing (LiteBlocks):** The core consists of 4 sequential residual blocks. Each *LiteBlock* passes data through a convolution, a non-linear GELU activation, and a second convolution with a residual connection to stabilize training and preserve fine details.
3. **Reconstruction:** A final 3×3 convolution compresses the refined 32-channel feature map back to a 3-channel shadow-free RGB image.

2.2. Addressing Technical Challenges

The model is specifically designed to address common shadow removal issues:

- **Penumbra Blending:** Using residual learning to smoothly blend soft shadow edges rather than just brightening dark areas.

- **Texture Preservation:** Mitigating texture loss and noise amplification in shadowed regions through GELU-activated convolutional blocks.
- **Object Discrimination:** Reducing the risk of altering naturally dark objects by focusing on spatially varying illumination changes.

2.3. Training Strategy

- **Dataset:** NTIRE shadow removal dataset containing 1,000 pairs of shadow and shadow-free (ground truth) images at 960×720 resolution.
- **Loss Function:** L1 Loss (Mean Absolute Error) to minimize pixel-wise reconstruction errors and maintain sharpness.
- **Optimizer:** AdamW with a learning rate of 2×10^{-4} .
- **Training Config:** Batch size of 1 for high-resolution stability, 20 epochs, and Automatic Mixed Precision (AMP) for speed.

3. Experimental Results

The model was evaluated using PSNR, SSIM, and LPIPS metrics.

- **Validation PSNR:** 20.263 dB
- **Test PSNR:** 19.889 dB
- **Inference Speed:** ~ 0.43 seconds per image

4. Conclusion

LiteRestormer provides a stable, low-cost solution for shadow removal. While a performance gap exists compared to complex transformer models, the current architecture offers high efficiency suitable for real-time applications like autonomous driving or robotic perception.

5. References

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